Unsupervised Recurrent Hyperspectral Imagery Super-Resolution Using Pixel-Aware Refinement

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***Abstract*— Unsupervised fusion-based hyperspectral imagery (HSI) super-resolution (SR) is an essential task of HSI processing, which aims to reconstruct a high-resolution (HR) HSI using only an observed low-resolution HSI and a conventional HR image. Although a large number of unsupervised HSI SR methods have been proposed, the heuristic handcrafted image priors adopted by the majority of these methods restrict their capacity to capture specific characteristics of the HSI, as well as their ability to generalize to noisy observation images. In this study, we investigate a fusion-based HSI SR framework with the deep image prior, in which the deep neural network (rather than a heuristic handcrafted image prior) is exploited to capture plenty of image statistics. Within this framework, we further propose an unsupervised recurrence-based HSI SR method using pixel-aware refinement, which utilizes the intermediate reconstruction results to self-supervise unsupervised learning. Due to containing the information of the image-specific characteristic, the proposed method achieves better performance, in terms of both accuracy and robustness to noise, compared with the existing methods. Extensive experiments on four HSI data sets demonstrate the effectiveness of the proposed method.**

***Index Terms*— Hyperspectral image super-resolution (SR), pixel-aware refinement, unsupervised deep learning.**

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I. Introduction

HYPERSPECTRAL imagery (HSI) is a category of images that contain both spectral information and spatial information [1]. Due to the benefits offered by its discriminative spectral information, HSI has been widely applied in remote sensing fields [2]–[4] and numerous computer vision tasks, such as environmental monitoring [5], scene classification [6]–[8], and target detection [9]–[11]. However, due to physical limitations, the captured HSIs usually have a high spectral resolution but a low spatial resolution, which restricts their use in many applications.

To address this problem, HSI super-resolution (SR) methods have been proposed enabling the reconstruction of HSIs with high resolution (HR) in both the spatial and spectral domain. Depending on the type of input images, existing HSI SR methods can be roughly divided into single image-based methods and fusion-based ones. Single image-based SR methods reconstruct the HR HSI by directly improving the spatial resolution of low-resolution (LR) HSI [12]–[17], while fusion-based methods utilize both LR HSI and conventional HR images (e.g., panchromatic images [18]–[21], multispectral images (MSIs) [22], [23], and RGB images [24], [25]) for HSI SR [26]–[30]. Compared with the single image-based SR methods, fusion-based methods can enlarge the spatial resolution of LR HSI to a far greater extent (i.e., a larger upsampling scale) since conventional HR image can provide sufficient spatial statistics for the desired HR HSI. Accordingly, fusion-based methods have been the primary focus of HSI SR-related research in recent years. In this study, we mainly focus on the fusion-based HSI SR method.

As noted earlier, fusion-based HSI SR methods aim at generating an HR HSI from an observed LR HSI and a conventional HR image. Depending on the approach utilized, fusion-based methods can be further subdivided into conventional fusion-based HSI SR methods and deep learning-based ones. In conventional fusion-based HSI SR methods, the LR HSI and conventional HR images are regarded as two degenerated images from the desired latent HR HSI in the spatial and spectral domains, respectively. Thus, these methods focus on how to inversely solve an HR HSI from the observed LR HSI and conventional HR image. Considering the ill-posed nature of this approach, image priors are always modeled to constrain the problem. Different heuristic handcrafted image

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priors, such as sparsity [29], [31], [32] and low-rank [24], [33], [34] priors, are consequently proposed (the details of these methods are presented in Section II). Despite being effective at regularizing the optimization problem, heuristic image priors show limitations when attempts are made to generalize them to challenging optimization problems. The reason can be attributed to the heuristic assumption of the prior and the restricted expressiveness capacity of the shallow structure, which impedes the further improvement of conventional fusion-based HSI SR methods.

Unlike the conventional fusion-based methods discussed earlier, most existing deep learning-based HSI fusion methods directly model the mapping function (i.e., relationship) between the HR HSI and the observed images, taking advantage of the powerful learning and expression ability of deep neural networks [35], [36]. A range of different network architectures (especially convolutional neural networks) is utilized, such as mapping functions. However, most existing deep learning-based methods are supervised methods and, accordingly, require large amounts of additional paired images (i.e., providing the latent HR HSI, LR HSI, and conventional HR images simultaneously for training [23], [37]), which is always infeasible in practical applications. Thus, recent researchers have begun to shift their focus to unsupervised deep learning-based methods [27], [38], which obtains the HR HSI from the observed LR HSI and conventional HR image only. Further details about deep learning-based methods are provided in Section II.

In this study, we explore an unsupervised deep learning-based HSI fusion method. Unlike most existing deep learning-based methods, which directly model the mapping function between the HR HSI and the observed images, our approach first utilizes a deep neural network as the image prior in order to capture plenty of expressive image statistics from the observed images [39], which are then exploited to constrain the inverse solving problem for HSI fusion. By doing this, the proposed method builds a connection between the deep learning-based method and the conventional traditional optimization-based method. Within this framework, we further propose an unsupervised recurrence-based HSI SR method using pixel-aware refinement, which utilizes the intermediate reconstruction results to self-supervise unsupervised learning. More specifically, we propose to recurrently update the input of the deep image prior. In addition, we propose to utilize the intermediate results in order to first measure the difficulties of reconstructing the pixels and then exploit this difficulty degree to measure how a certain pixel has recovered. As the proposed method utilizes information regarding image-specific characteristics, it is more robust compared with both conventional fusion-based HSI SR method and deep learning-based HSI SR one.

In summary, the main contributions of the proposed method are as follows: 1) instead of utilizing the heuristic handcrafted image prior, we propose an unsupervised fusion-based HSI SR framework with the deep image prior, which has better capacity to regularize the ill-posed HSI inverse solving problem; 2) within the deep image prior-based framework, a recurrent strategy-based method with the pixel-aware refinement that is self-supervised by the intermediate results is further proposed to boost performance; and 3) experimental results on four data sets demonstrate that the proposed method achieves better performance in terms of reconstruction accuracy, as well as robustness to noise, compared with other state-of-the-art fusion-based HSI SR methods.

The remainder of this article is organized as follows. Section II introduces the background and related works of this study. Section III presents the proposed fusion SR framework with the deep image prior, along with the proposed recurrent fusion-based HSI SR method using pixel-aware refinement. Section IV summarizes the experimental results on four data sets. Finally, Section V concludes this article.

II. Background and Related Works

In this study, we denote the target HR HSI as $\mathcal{Z} \in \mathbb{R}^{B\times P\times Q}$ with P, Q, and B representing its height, width, and band number, respectively. $\mathcal{X} \in \mathbb{R}^{B\times p\times q}$ is the observed LR HSI, with p, q, and B denoting its height, width, and band number ($p \ll P$ and $q \ll Q$). $\mathcal{Y} \in \mathbb{R}^{b\times P\times Q}$ is the observed conventional HR image (e.g., MSI and RGB image), where b is its band number ($b \ll B$). For more convenient representation, we rearrange $\mathcal{Z} \in \mathbb{R}^{B\times P\times Q}$, $\mathcal{X} \in \mathbb{R}^{B\times p\times q}$, and $\mathcal{Y} \in \mathbb{R}^{b\times P\times Q}$ into matrices as $\mathbf{Z} \in \mathbb{R}^{B\times N}$, $\mathbf{X} \in \mathbb{R}^{B\times n}$, and $\mathbf{Y} \in \mathbb{R}^{b\times N}$, where $n = p \times q$ and $N = P \times Q$. In these matrices, each row denotes the reflectance values of all pixels in one specific band, while each column denotes the spectrum at one pixel in the spectral dimension.

Fusion-based HSI SR methods aim to reconstruct the latent HR HSI $\mathbf{Z}$ from the observed LR HSI $\mathbf{X}$ and the conventional HR image $\mathbf{Y}$. Based on the fact that the LR HSI and conventional HR image are two degenerated images from the desired latent HR HSI in the spatial and spectral domains, respectively, the relationships between $\mathbf{Z}$, $\mathbf{X}$, and $\mathbf{Y}$ can be represented as

$$\mathbf{X} = \Phi(\mathbf{Z}) + \mathbf{N}_{\mathbf{X}}, \quad \mathbf{Y} = \Psi(\mathbf{Z}) + \mathbf{N}_{\mathbf{Y}} \tag{1}$$

where $\Phi(\cdot)$ and $\Psi(\cdot)$ denote the spatial and spectral degeneration functions, respectively, and $\mathbf{N}_{\mathbf{X}}$ and $\mathbf{N}_{\mathbf{Y}}$ are the noise in the observed images.

Different kinds of fusion-based HSI SR methods have been proposed based on the observation model presented in formula (1). These methods can be roughly divided into conventional fusion-based and deep learning-based ones. In the following, we will separately introduce these two kinds of methods and their related works.

A. Conventional Fusion-Based HSI SR Methods

Conventional HSI SR methods [24], [26], [29], [31]–[34] focus on how to inversely solve a latent HR HSI from the observed LR HSI and conventional HR image. This can be framed as an optimization problem as

$$\min_{\mathbf{Z}} \ \|\mathbf{X} - \Phi(\mathbf{Z})\|_F^2 + \mu\|\mathbf{Y} - \Psi(\mathbf{Z})\|_F^2 + \lambda\Gamma(\mathbf{Z}) \tag{2}$$

in which $\|\cdot\|_F$ denotes the Frobenius norm and μ is a balance weight. Due to the ill-posed nature of this problem, image prior $\Gamma(\mathbf{Z})$ (reflecting the characteristics of the image) is exploited to constrain the abovementioned optimization problem. Different kinds of priors are exploited

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as $\Gamma(\mathbf{Z})$ to regularize formula (2). For example, Lanaras *et al.* [26] assumed that the latent HR HSI can be decomposed into a sparse abundance matrix together with a nonnegative endmember matrix based on the linear mixed model and then further propose to utilize the sparse characteristic as the image prior in order to reconstruct the latent HR HSI. Based on coupled nonnegative matrix factorization (CNMF) framework [20], Yang *et al.* [31] proposed further incorporating sparse and proximal regularizers to promote HSI SR performance. Dong *et al.* [29] proposed to reconstruct the HSI via the sparsity prior in the spectral domain, as well as nonlocal similarity characteristics in the spatial domain. Besides sparsity, low rank is another characteristic commonly used to regularize the abovementioned ill-posed problem. Wu *et al.* [34] proposed to jointly exploit the globally and locally low-rank characteristics in the spatial and spectral domains simultaneously for HSI SR. Zhang *et al.* [24] further exploited the clustering manifold structure of the latent HR HSI to improve reconstruction accuracy.

The advantage of conventional fusion-based methods is that they can estimate HR HSI in an unsupervised manner. However, the existing image priors (e.g., sparsity prior [29]–[32] or low-rank prior [24], [33], [34], [40]) show limitations to the challenging optimization problem. The reason can be attributed to the heuristic assumption of the prior, as well as the restricted expressiveness capacity of the shallow structure (especially when the SR scale factor is large, e.g., 8, 16, or 32 times), which impedes the further improvement of HSI SR methods.

B. Deep Learning-Based HSI SR Methods

In recent years, deep learning-based methods have demonstrated their superiority in image-processing-related tasks. This is due to their powerful representation ability, which has also promoted the development of deep learning-based HSI SR methods. Unlike the conventional fusion-based HSI SR methods that inversely solve the HR HSI, deep learning-based HSI SR methods [16], [21], [22], [27], [28], [37] directly build the relationship (i.e., mapping function) between the HR HSI and the observed LR HSI, as well as the conventional HR image. Thus, different kinds of deep learning-based methods (particularly convolutional neural network-based methods) are exploited to depict this relationship. For example, Palsson *et al.* [22] proposed utilizing a 3-D convolutional network to construct the backbone of an HSI SR network, while Scarpa *et al.* [41] developed a compact residual convolutional network to model such a relationship. Aiming to further utilize the intrinsic correlations between HSI and MSI, Xie *et al.* [23] proposed a uniform network, which further considers the low-rankness characteristics of spectra in the latent HR HSI when modeling the deep learning network. Li *et al.* [21] first trained a deep convolutional neural network for HSI SR using large numbers of labeled RGB images and then fuse the spatial and spectral details into a reconstructed HR HSI with a guided image filter and a gain matrix.

Although modeling a supervised end-to-end HSI SR network is the most direct way of utilizing deep learning technology, this approach is outside the scope of this study since such a supervised network demands the availability of many additional pair training samples (including latent HR HSIs, observed LR HSIs, and conventional HR images). However, in real applications such as remote sensing, it often difficult to collect sufficient and satisfactory HR HSIs for training. Thus, modeling an unsupervised deep learning-based method that is trained only on the observed LR HSI and conventional HR image is more desirable in real-world applications and has, thus, also become the focus of research into deep learning-based methods in very recent years. Several unsupervised deep methods have been newly proposed for the HSI SR context [27], [38]. For example, by utilizing two coupled encoder–decoder networks to separately extract the spectra and corresponding proportional coefficients from the input LR HSI and HR MSI, Qu *et al.* [38] proposed a state-of-the-art unsupervised deep SR method, named uSDN. uSDN encourages the representation of the input HSI and MSI to follow a sparse Dirichlet distribution and optimizes the angle difference during the optimization process, which makes uSDN well restrain the reconstructed HSI and reduce the spectral distortion. However, in a departure from the existing unsupervised HSI fusion deep method that models the relationship between the HR HSI and the observed LR HSI/conventional HR image directly, we utilize a deep neural network to model the image prior and then inversely obtain the desired HR HSI.

In addition, most of the abovementioned methods focus primarily on establishing a new or more powerful SR model on clean observation data, rather than on specifically designing a model that can appropriately eliminate the negative influence of noise accompanied with the observed images. In applications, these methods [23]–[25], [29], [38] often utilize clean images or a fixed high signal-to-noise ratio (SNR) (e.g., 30 dB or higher SNR) images as the observations in the experiments. As a result of this practice, when confronted with different levels of noise corruption in the observed images, the performance of these existing fusion-based HSI SR methods tends to decline significantly. By contrast, the proposed method not only presents a new fusion-based HSI SR framework but also exhibits strong robustness over different levels of noise corruption in our experiments.

III. Proposed Method

To take advantage of the unsupervised nature of the conventional fusion-based HSI SR methods, as well as the powerful representation ability of deep neural networks, we propose to exploit deep neural networks to model the image prior in this study. Recently, the concept of the deep image prior was proposed in [39] to utilize a generator network to depict an image prior. However, such an image prior has not yet been explored in a fusion-based HSI SR problem. Inspired by this, we first introduce the deep image prior directly into the fusion-based HSI SR problem in this study in Section III-A and then propose a new deep image prior-based HSI SR method in Section III-B.

A. Unsupervised HSI SR Framework With Deep Image Prior

As discussed in Section II, existing image priors (e.g. sparse prior or low-rank prior) show limitations in terms of their

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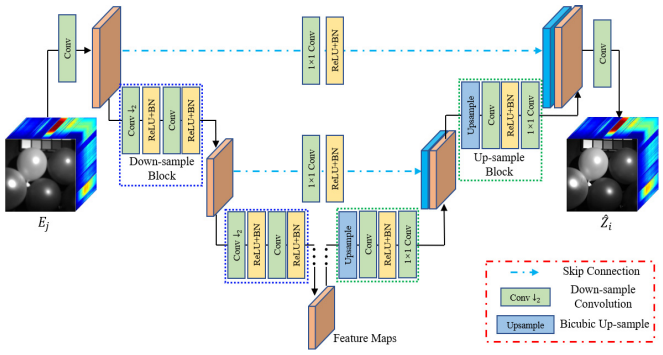


Fig. 1. Structure of the generator network constructed in this study.

ability to generalize to challenging optimization problems (due to the heuristic assumption and the restricted expressiveness capacity of the shallow structure for the conventional fusion-based HSI SR methods). By taking advantage of powerful representation ability of deep neural networks, we can replace the heuristic prior $\Gamma(\mathbf{Z})$ in formula (2) with a deep neural network f_Θ as

$$\min_{\Theta} \ \|\mathbf{X} - \Phi(\mathbf{Z})\|^2 + \mu\|\mathbf{Y} - \Psi(\mathbf{Z})\|^2 \quad \text{s.t. } \mathbf{Z} = f_\Theta(\mathbf{E}) \tag{3}$$

where Θ and $\mathbf{E}$ represent the parameters and input of the constructed deep neural network, respectively. Due to the fact that a well-designed generator network structure has the capability to capture image characteristics [39], we model f_Θ using a generator network structure to capture the abundant image characteristics of $\mathbf{Z}$. It can be seen that, in this framework, fusion-based HSI SR is cast into optimizing the generator network f_Θ, which is an unsupervised deep learning HSI SR method that depends only on the observed images $\mathbf{X}$ and $\mathbf{Y}$.

Considering that a 3-D convolution network contains more parameters than the 2-D convolution alternative, training a 3-D convolution network requires more training data and consumes more training time to archive convergence. However, in the unsupervised framework proposed in this work, the useful information in the observed images is extremely limited. Thus, using a 3-D convolution network as the generator network increases the difficulty for training. For this reason, we utilize a 2-D convolution network as the generator inspired by [39] in this study. The structure of the constructed generator network is illustrated in Fig. 1. More specifically, the generator network consists primarily of several downsample and upsample blocks. The downsample block enables the generator network to have a large receptive field, while the upsample block can enlarge the spatial size of feature maps. Meanwhile, the downsample and upsample blocks that are located at the same level will be connected by a skip connection, which is implemented by an individual convolution layer, a rectified linear unit (ReLU), and a batch normalization (BN) layer.

B. Recurrent Fusion-Based HSI SR Method Using Pixel-Aware Refinement

While f_Θ constructed in formula (3) has better representation ability compared with $\Gamma(\mathbf{Z})$ formulated in formula (2), it is crucial to optimize formula (3), so as to obtain a good HSI SR performance. Adopting uniform noise as input is simple and can be applied to any images [39]; however, this approach neglects the image-specific information (i.e., taking noise as input does not consider the differences among different images).

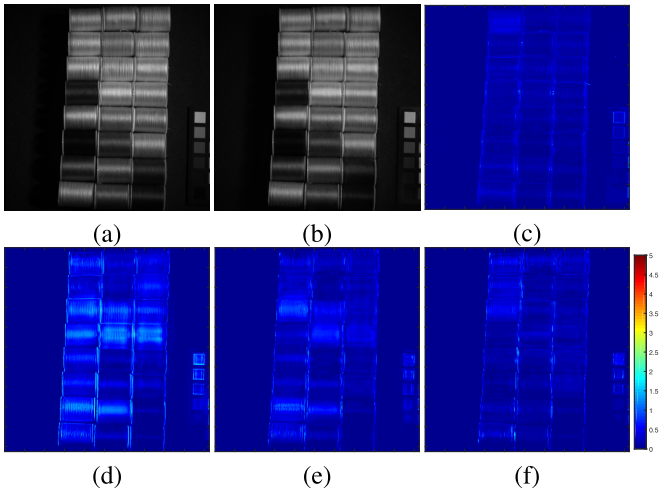


Fig. 2. Reconstruction results, together with the different maps on the "Thread_spools" HSI in the CAVE data set. (a) Ground Truth. (b) final reconstruction result of the proposed method (Ours). (c) Error map between (a) and (b). (d)–(f) Difference maps obtained from the 300th, 400th, and 500th intermediate reconstruction results, respectively. A darker color in the difference maps indicates a smaller difference. (d) Diff_Map3. (e) Diff_Map4. (f) Diff_Map5.

Considering that Formula (3) can be optimized using an iterative strategy, the reconstruction results are improved gradually (i.e., the reconstruction results improve as the number of iterations increases). Thus, the intermediate reconstruction results provide a clue regarding how regularization can be performed. Inspired by the self-supervised strategy, we propose utilizing the intermediate results to guide the optimization problem in order to obtain a more reliable and accurate HSI fusion-based SR method. For better clarification, an example is illustrated in Fig. 2, in which the difference maps between the intermediate reconstructed images on the "Thread_spools" HSI in the CAVE data set are given in Fig. 2(d)–(f). A darker color in the difference maps indicates a smaller difference. We can see that the background pixels are easier to reconstruct than those from the foreground, as well as the details. The differences of the pixels from the background approach the smallest value (i.e., 0) in the early stage (i.e., before the 300th epoch) and remain invariant from the 300th epoch. By contrast, the differences of the pixels from the foreground and the details vary from the 300th epoch to the 500th epoch.

Thus, it is necessary to focus on how to effectively incorporate the intermediate results into formula (3). For this purpose, we first utilize the upsampled image as the initial $\mathbf{E}$ and then update this $\mathbf{E}$ with the intermediate reconstruction result, which is an image-specific item. Subsequently, we propose utilizing the intermediate results to measure the difficulties of reconstructing the pixels and then exploit this degree of difficulty to measure how a certain pixel is recovered. For clarity, we can mathematically represent the above descriptions

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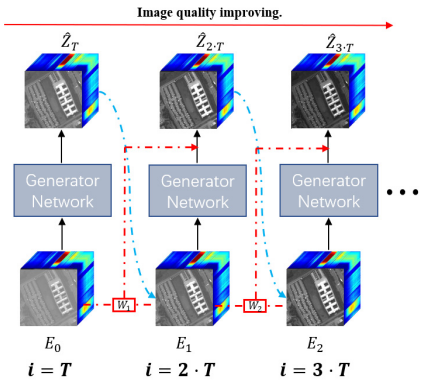


Fig. 3. Flowchart of the proposed recurrent model with pixel-aware refinement mechanism. The dotted lines in blue represent how we update the input $\mathbf{E}_j$ in stages, while the dotted lines in red represent how we update the pixel-aware weights in stages.

as

$$\min_{\Theta} \mathbf{W}_j^H \odot \|\mathbf{X} - \Phi(\hat{\mathbf{Z}}_i)\|^2 + \mathbf{W}_j^M \odot \|\mathbf{Y} - \Psi(\hat{\mathbf{Z}}_i)\|^2$$
$$\text{s.t. } \hat{\mathbf{Z}}_i = f_\Theta(\mathbf{E}_j), \quad \mathbf{E}_j = \hat{\mathbf{Z}}_{T\times j}, \quad j = \lfloor i/T \rfloor \tag{4}$$

where $\hat{\mathbf{Z}}_i$ represents the reconstructed HR HSI at the ith epoch, while $\mathbf{E}_j$ is the input of the generator network at the jth stage (one stage consists of T epochs). $\mathbf{W}_j^H$ and $\mathbf{W}_j^M$ refer to the pixel-aware weights of LR HSI $\mathbf{X}$ and HR MSI $\mathbf{Y}$, respectively, at the jth stage. $\odot$ denotes the elementwise multiplication, T represents the number of epochs within one stage, and $\lfloor i/T \rfloor$ represents the round-down operation on i/T.

It can be seen from formula (4) that the input of the generator network is related to the intermediate reconstruction results. It is notable that we update $\mathbf{E}_j$ at every stage with the latest reconstruction result obtained at the $T \times j$th epoch (i.e., $\mathbf{E}_j = \hat{\mathbf{Z}}_{T\times j}$), which results in the proposed method being a recurrent fusion-based model. Fig. 3 illustrates the recurrent updating flowchart, in which the dotted blue lines represent how we update the input $\mathbf{E}_j$ in stages. The advantage of updating $\mathbf{E}_j$ at every stage is that the image-specific information contained in the input helps to narrow the discrepancy between the input and target images of the generator network. As a result, it accelerates the convergence speed of the proposed method (see the ablation study for more details).

The other important issue is the pixel-aware weights $\mathbf{W}_j^H$ and $\mathbf{W}_j^M$, which reflects the degree of difficulty associated with reconstructing different pixels. As discussed earlier, the results of the reconstruction gradually improve, which prompts us to determine those weights with the intermediate reconstruction results. Since the reconstruction results are obtained for the latent HR HSI, we first determine the pixel-aware weights $\mathbf{W}_j$ at the jth stage with the intermediate reconstruction results obtained from neighboring stages as

$$\mathbf{W}_j = \text{Exp}(\alpha \cdot |\mathbf{E}_j - \mathbf{E}_{j-1}|) \quad \text{s.t. } j = \lfloor i/T \rfloor \tag{5}$$

where α is the weight coefficient and Exp indicates the exponential function. We then determine the pixel-aware weights $\mathbf{W}_j^H$ and $\mathbf{W}_j^M$ as

$$\mathbf{W}_j^H = \Phi(\mathbf{W}_j), \quad \mathbf{W}_j^M = \mu \times \Psi(\mathbf{W}_j) \tag{6}$$

Algorithm 1 Proposed Unsupervised Recurrent HSI SR Method Using Pixel-Aware Refinement

Input: LR HSI $\mathbf{X}$, HR MSI $\mathbf{Y}$.
Initialization: Initialize $\mathbf{E}_0$ with the concatenation of upsampled $\mathbf{X}$ and $\mathbf{Y}$. $\mathbf{W}_0^H = \mathbf{1}$, $\mathbf{W}_0^M = \mathbf{1}$, the maximum iteration number $I = 2000$, epoch $i = 1$, stage $j = 0$, interval $T = 100$, weight coefficient $\alpha = 2.5$.
for $i<I$ **do**
  Reconstruct the latent HR HSI $\hat{\mathbf{Z}}_i$ via optimizing the Formula (4) with the current input $\mathbf{E}_j$.
  if $i/T == 0$ **then**
    $j = j + 1$;
    Update the input $\mathbf{E}_j$ with the latest reconstructed $\hat{\mathbf{Z}}_i$;
    Calculate the pixel-aware weight $\mathbf{W}_j^H$ and $\mathbf{W}_j^M$ according to Formula (5) and Formula (6).
Return the reconstructed latent HR HSI $\hat{\mathbf{Z}}_I$

after which the obtained $\mathbf{W}_j^H$ and $\mathbf{W}_j^M$ are then applied to formula (4) for optimization. Considering that these weights reflect the degree of difficulty associated with reconstructing different pixels, optimization formula (4) will first reconstruct the pixels that are easy to reconstruct (such as those in background) and then turn to reconstructing those that are more difficult, including details and noise. The red dotted lines in Fig. 3 represent how we update the pixel-aware weights.

In the above description, $\Phi(\cdot)$ and $\Psi(\cdot)$ indicate a general form without giving specific representations. To obtain a fully HSI-oriented SR network, we provide the detailed modeling of $\Phi(\cdot)$ and $\Psi(\cdot)$ in the following. Considering that the degeneration $\Phi(\cdot)$ in the spatial domain can be simply modeled via a blur operation together with a downsampling operation, we formulate $\Phi(\mathbf{Z})$ as

$$\Phi(\mathbf{Z}) = (k * \mathbf{Z}) \downarrow_s \tag{7}$$

where k denotes the blur kernel and $\downarrow_s$ indicates the downsampling operation with SR scale factor s. As for the degeneration $\Psi(\cdot)$ in the spectral domain, it can be formulated as

$$\Psi(\mathbf{Z}) = \mathbf{P}\mathbf{Z} \tag{8}$$

where $\mathbf{P}$ denotes a linear mapping matrix, which reflects the spectral response function and depends on the corresponding HSI camera. Once the HSI camera is given, $\mathbf{P}$ is fixed. With the above definitions, we can simply utilize a convolution layer with a predefined kernel and a fully connected layer to implement $\Phi(\cdot)$ and $\Psi(\cdot)$, respectively.

Using the symbols and representations defined earlier, we summarize the procedure of the proposed recurrent fusion-based HSI SR method using pixel-aware refinement in Algorithm 1 and further present a flowchart of the procedure in Fig. 4.

IV. Experimental Results and Analysis

A. Experimental Settings

1) Data Sets: To demonstrate the effectiveness of the proposed method, we conduct experiments on three benchmark

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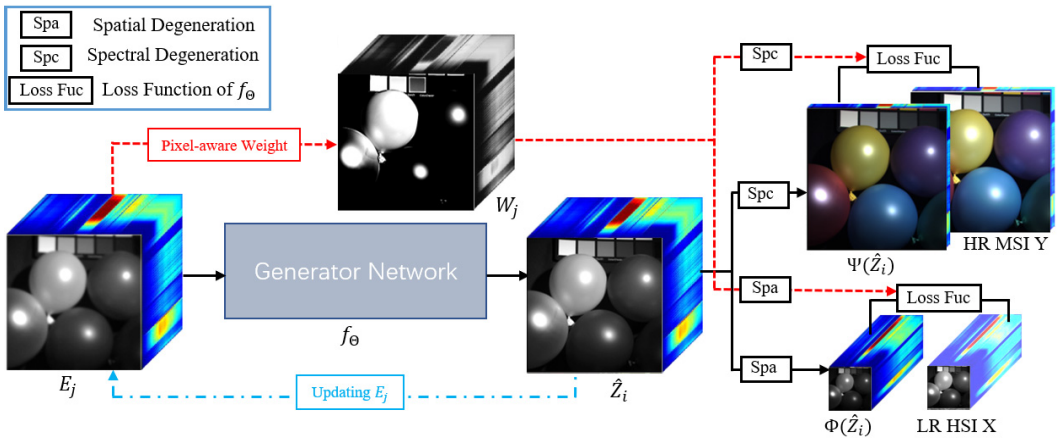


Fig. 4. Flowchart of the proposed method. Here, solid arrows in black indicate the main HSI SR process, while the dotted arrows in blue and red represent the proposed recurrent pixel-aware refinement mechanism.

HSI data sets, including CAVE [42], Chikusei [43], and Washington DC Mall (Wdc), as well as one real HSI data set HypSen [44].

The CAVE data set contains 32 indoor HSIs. Each HSI is of size 512×512 in the spatial domain and has 31 bands covering the spectrum from 400 to 700 nm with a 10-nm interval. The Chikusei data set was captured by the Headwall Hyperspec-VNIR-C imaging sensor over agricultural and urban areas in Chikusei, Japan, on July 29, 2014. It contains 128 bands, ranging from 363 to 1018 nm, and its spatial size is 2517×2335. In the experiments, we crop 13 nonoverlapping subimages from the original Chikusei data set, and the resulted subimages are of size 256×256 in the spatial domain. We then utilize these 13 subimages for experiments. The Wdc data set contains HSIs with 210 bands, covering the spectral wavelength from 0.4 to 2.5 μm. After removing the water-absorption bands, we retain a total of 191 bands and crop a 200×200 spatial-size subimage centered at the original 1280×307 image for experiments.

The HypSen data set contains both HSI and MSIs of the same scene in Lafayette, LA, USA, which was captured separately in October and November 2015. The HSI is collected by the Hyperion sensor carried on the Earth Observing-1 (EO-1) satellite with 30-m spatial resolution. It has 242 spectral bands, covering the spectral range from 400 to 2500 nm. The MSI is captured via the Sentinel-2A satellite with 10-m spatial resolution, and the collected image contains six bands. To assess the effectiveness of all methods under real conditions in experiments, a 100×100 spatial-size subimage is cropped from the original HSI, while a 300×300 spatial-size subimage is also cropped from the corresponding MSI.

2) Implementation Details: In experiments, three benchmark data sets (CAVE, Chikusei, and Wdc) are adopted as the simulated data. Each adopted HSI in these three data sets is regarded as the ground truth of the latent HR HSI $\mathbf{Z}$, while the corresponding LR HSI $\mathbf{X}$ and HR MSI image $\mathbf{Y}$ are downsampled from $\mathbf{Z}$. More specifically, a zero-mean Gaussian blur kernel of size 8×8 is convolved with $\mathbf{Z}$ to generate $\mathbf{X}$, as in formula (7). As for the HR MSI image, we follow the settings outlined in [24] and [29], utilize the corresponding response function of each data set as $\mathbf{P}$, and then obtain the HR MSI image $\mathbf{Y}$, as in formula (8). The performance on these three data sets can be assessed by comparing the results between the reconstructed HSI and the ground-truth HR HSI $\mathbf{Z}$. To demonstrate the effectiveness of the proposed method for HSI SR, we conduct extensive experiments on three benchmark HSI data sets with three SR scale factors, specifically 8, 16, and 32 times (i.e., the spatial resolution of the LR HSI is enlarged by eight times, 16 times, or 32 times). In addition, to verify the performance of HSI SR when the observed images are noisy, we separately add Gaussian noise of different intensities into $\mathbf{X}$ and $\mathbf{Y}$ to obtain the noisy observation images, as suggested in previous HSI denoising works.

Moreover, since the HypSen data set contains both HSI and MSI, it is adopted to verify the performance of HSI SR methods in real applications. More specifically, we treat the HSI and MSI data in HypSen as the LR HSI $\mathbf{X}$ and HR MSI $\mathbf{Y}$, respectively, with which we reconstruct the HR HSI and then visually compare the results obtained from different methods.

In this study, we utilize three downsample blocks followed by three upsample blocks to construct the generator network. The Adam algorithm is applied to the generator network for optimization. The learning rate is set as $1.2e-3$ and halved at the 1000th epochs. The maximum number of iterations I is set as 2000. In addition, we set the interval T (the maximum number of epochs within one stage) as 100, meaning that we update the input $\mathbf{E}_j$ and pixel-aware weights every 100 epochs. Finally, the weight a, which is utilized to determine the pixel-aware weights, is set as 2.5.

3) Competitors and Evaluation Metrics: Considering that the proposed method is an unsupervised method, we compare it with four existing state-of-the-art unsupervised fusion-based HSI SR methods, including CMS [24], NSSR [29], Yong [25], and uSDN [38]. Among these four competing methods, uSDN is an unsupervised deep learning-based method. To quantitatively measure the performance of the involved methods, we adopt four evaluation metrics, including root mean square

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TABLE I

PERFORMANCE OF BASELINE AND OURS ON THE CAVE DATA SET. THE SR SCALE FACTOR s IS SET AS 8, AND THE OBSERVED IMAGES (LR HSI AND HR MSI) ARE CONTAMINATED BY NOISE WITH DIFFERENT LEVELS OF INTENSITY. THE BEST RESULTS ARE IN BOLD

SNR	Methods	RMSE	PSNR	SAM	SSIM
HSI 30dB MSI 30dB	Baseline	2.79	39.79	7.53	0.9637
	Ours	**2.17**	**41.71**	**4.49**	**0.9812**
HSI 30dB MSI 25dB	Baseline	2.88	39.48	7.52	0.9601
	Ours	**2.62**	**40.14**	**5.22**	**0.9653**
HSI 25dB MSI 30dB	Baseline	3.73	37.43	11.95	0.9306
	Ours	**2.58**	**40.51**	**5.39**	**0.9757**

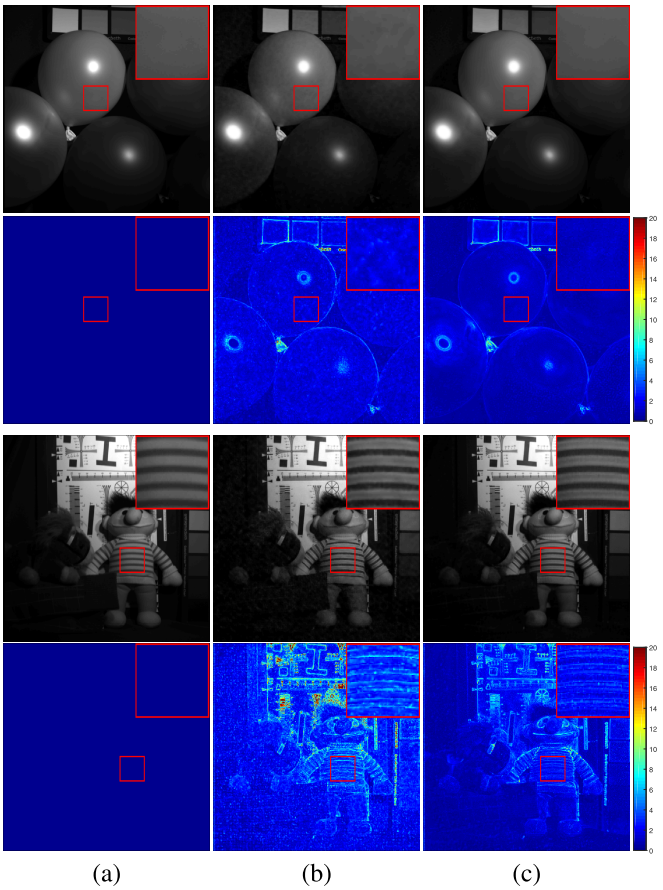


Fig. 5. Reconstruction results of Baseline and Ours on the HSIs "balloons" and "chart_and_stuffed_toy" from the CAVE data set. The first and third rows give the reconstructed images of the 31st band, while the second and fourth rows present the reconstruction error maps. The SR scale factor s is set as 8, and the SNRs of the observed LR HSI and HR MSI are 25 and 30 dB, respectively. GT denotes the ground-truth images. (a) GT. (b) Baseline. (c) Ours.

error (RMSE), peak SNR (PSNR), spectral angle mapper (SAM), and structural similarity index (SSIM). RMSE and PSNR calculate the average numerical differences between the reconstructed HR HSI and the ground truth, while SAM measures the spectral distortion between the reconstructed HR HSI and the ground truth. SSIM computes the structural similarity between the reconstructed HSI and the reference image in the spatial domain. To facilitate fair comparison,

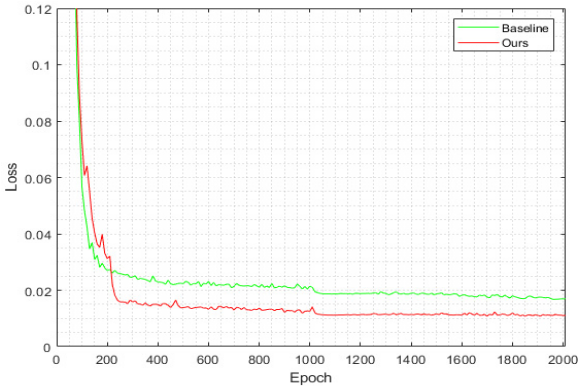


Fig. 6. Variations of loss with an increasing number of iterations from Baseline and Ours. Only the LR HSI is contaminated by noise, and the corresponding SNR is 30 dB.

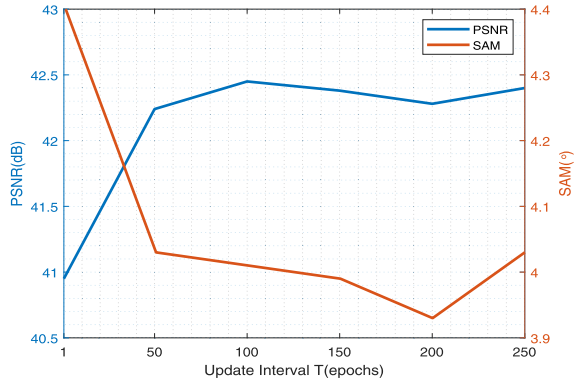


Fig. 7. Variations of PSNR and SAM from Ours with T. Only the LR HSI is contaminated by noise, and the corresponding SNR is 30 dB.

we follow the settings outlined in [24] and [29] to compute the RMSE on the 8-bit image. In experiments, smaller values of RMSE and SAM and larger values of PSNR and SSIM indicate better reconstruction results.

B. Ablation Study

Within the proposed method, the pixel-aware refinement mechanism, the interval T, and the weight coefficient α will influence the performance. To testify how they affect the performance of the proposed method, the following ablation studies were conducted on the CAVE data set.

1) Recurrent Pixel-Aware Refinement Mechanism: In this experiment, we compare the reconstruction results obtained by optimizing formula (3) (referred to as Baseline) and formula (4) (referred to as Ours). As noted above, Baseline is the method that directly introduces the deep image prior framework for HSI SR and adopts the default settings (i.e., uniform noise for $\mathbf{E}$) without considering the image-specific characteristics of the HR HSI. The only difference between Baseline and Ours lies in the recurrent pixel-aware refinement mechanism. For a fair comparison, we adopt the same settings for these two methods. The experiments are conducted on the CAVE data set when the SR scale factor s is set as 8 (i.e., we enlarge the LR HSI

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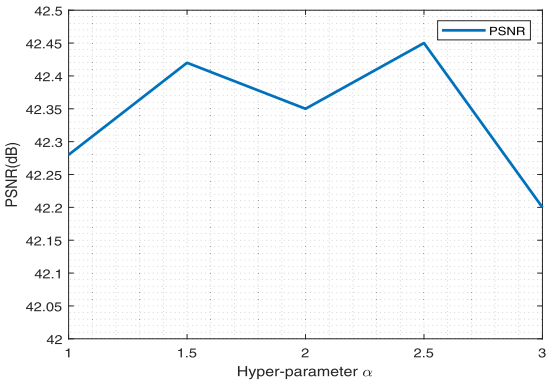


Fig. 8. Variations of PSNR from Ours with α. The SNR of LR HSI is 30 dB.

TABLE II

PERFORMANCE OF EACH METHOD ON THE CAVE DATA SET. THE SR SCALE FACTOR IS SET AS 8, AND THE LR HSI IS CONTAMINATED BY NOISE WITH DIFFERENT INTENSITY. THE BEST RESULTS ARE IN BOLD

SNR	Methods	RMSE	PSNR	SAM	SSIM
30dB	Yong	5.81	37.84	7.60	0.9574
	NSSR	2.13	41.96	8.09	0.9700
	CMS	2.03	42.28	8.19	0.9712
	uSDN	4.21	36.48	15.26	0.9191
	Ours	**1.98**	**42.45**	**4.01**	**0.9867**
25dB	Yong	2.93	39.15	8.38	0.9688
	NSSR	2.98	39.21	13.00	0.9247
	CMS	2.73	39.83	12.46	0.9341
	uSDN	4.43	36.06	14.40	0.9114
	Ours	**2.18**	**41.73**	**4.46**	**0.9851**
20dB	Yong	3.18	38.42	10.27	0.9579
	NSSR	4.85	35.15	20.47	0.8149
	CMS	4.22	36.20	18.83	0.8501
	uSDN	4.53	35.86	15.55	0.9044
	Ours	**2.61**	**40.48**	**5.37**	**0.9804**

eight times in the spatial domain). Moreover, to assess how the performance of both methods is influenced by noise, different intensities of Gaussian noise are separately added into the observed images. The numerical results on the CAVE data set are summarized in Table I, in which the results are obtained by averaging the reconstruction results of each HSI in the CAVE data set. We can see from the results that Ours achieves better performance than Baseline on all measures, with any kind of noise intensity, which demonstrates that the recurrent pixel-aware refinement mechanism can obviously improve the performance of HSI SR in terms of both reconstruction accuracy and robustness to noise. For further clarification, we also visualize the HSI SR results of these two methods in Fig. 5, including the reconstruction images and the error maps. The darker color in the error map indicates better reconstruction results. From the reconstructed band images and the error maps, it can be seen that obvious noise still exist in the results from Baseline. On the contrary, the results from Ours is cleaner and more smooth than those from Baseline. Thus, we can conclude that the proposed method can adjust the

TABLE III

PERFORMANCE OF EACH METHOD ON THE CAVE DATA SET. THE SR SCALE FACTOR IS SET AS 8, AND THE HR MSI IS CONTAMINATED BY NOISE WITH DIFFERENT INTENSITY. THE BEST RESULTS ARE IN BOLD

SNR	Methods	RMSE	PSNR	SAM	SSIM
30dB	Yong	3.14	38.60	7.85	0.9583
	NSSR	2.25	41.43	9.46	0.9595
	CMS	2.46	40.66	10.68	0.9488
	uSDN	4.21	36.24	14.30	0.9162
	Ours	**2.13**	**41.89**	**3.84**	**0.9827**
25dB	Yong	3.61	37.42	9.46	0.9257
	NSSR	3.22	38.46	14.49	0.9021
	CMS	3.67	37.36	16.20	**0.9776**
	uSDN	4.78	34.99	17.17	0.8695
	Ours	**2.30**	**41.17**	**4.16**	0.9732
20dB	Yong	4.70	35.23	12.29	0.8493
	NSSR	5.20	34.43	21.61	0.7778
	CMS	5.76	33.51	23.04	0.7519
	uSDN	6.28	32.66	21.12	0.7645
	Ours	**3.04**	**38.91**	**5.05**	**0.9390**

TABLE IV

PERFORMANCE OF EACH METHOD ON THE CAVE DATA SET. THE SR SCALE FACTOR IS SET AS 8, AND BOTH LR HSI AND HR MSI ARE CONTAMINATED BY NOISE WITH DIFFERENT INTENSITY. THE BEST RESULTS ARE IN BOLD

SNR	Methods	RMSE	PSNR	SAM	SSIM
HSI 30dB MSI 30dB	Yong	3.11	38.65	8.54	0.9554
	NSSR	2.57	40.37	11.42	0.9434
	CMS	2.77	39.70	12.83	0.9286
	uSDN	4.48	35.76	18.18	0.9001
	Ours	**2.17**	**41.71**	**4.49**	**0.9812**
HSI 30dB MSI 25dB	Yong	3.59	37.45	10.03	0.9195
	NSSR	3.42	37.96	15.61	0.8890
	CMS	3.86	36.92	17.47	0.8609
	uSDN	5.05	34.65	18.76	0.8453
	Ours	**2.62**	**40.14**	**5.22**	**0.9653**
HSI 25dB MSI 30dB	Yong	3.19	38.42	9.32	0.9515
	NSSR	3.19	38.61	14.44	0.9095
	CMS	3.33	38.17	15.72	**0.9824**
	uSDN	4.79	35.17	17.57	0.8750
	Ours	**2.58**	**40.51**	**5.39**	0.9757

deep image prior to better cope with the noise and reconstruct the details with the pixel-aware refinement.

In addition, we plot the variations of loss with an increasing number of iterations from Baseline and Ours in Fig. 6. It can be seen that, with the higher number of iterations, the loss of Ours reaches a lower value more easily than that of Baseline, which reflects that the proposed method has a faster convergence speed.

2) Interval T: Interval T is an important parameter, which controls the frequency of updating the pixel-aware weights and the input of the generator network. Fig. 7 shows the plots of the reconstruction results (including PSNR and SAM) of Ours as they vary with different values of T on the CAVE data set when the SR scale factor s is 8. It can be seen that the performance of the proposed method declines when T is too

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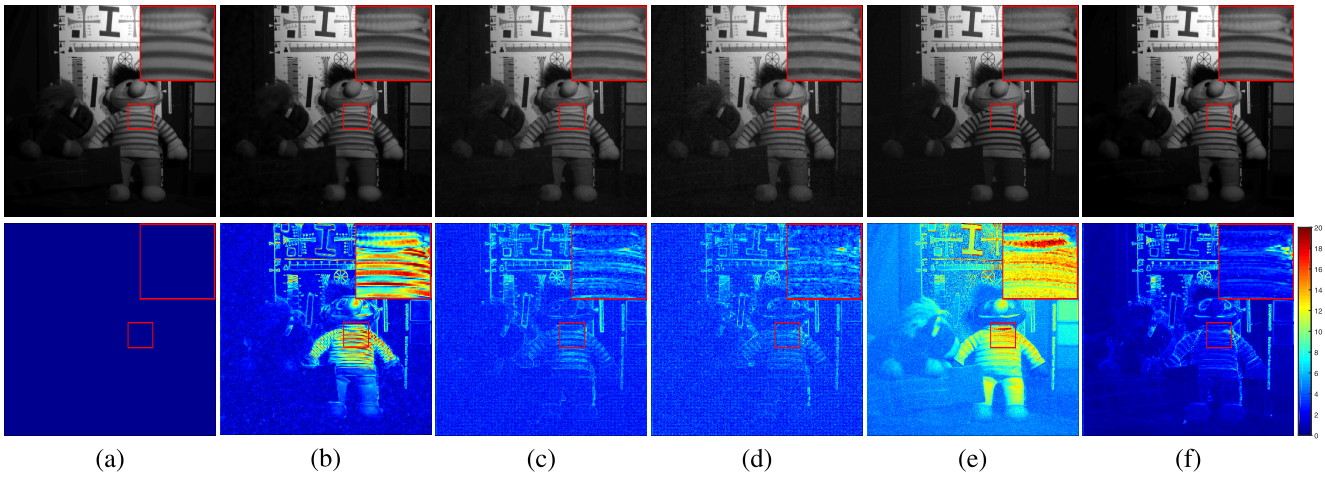


Fig. 9. Reconstruction results of HSI "chart_and_stuffed_toy" in the CAVE data set. The first row contains the reconstructed 31st band images from different methods, while the second row presents the reconstruction error maps. The SR scale factor is set as 8, and the SNRs of both the LR HSI and HR MSI are 30 dB. (a) GT. (b) Yong. (c) NSSR. (d) CMS. (e) uSDN. (f) Ours.

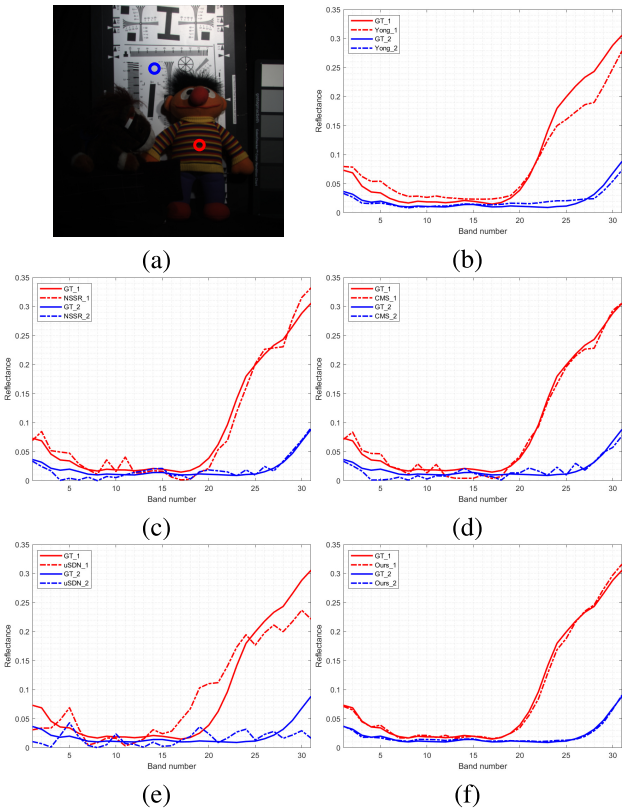


Fig. 10. Reconstructed spectral curve of two representative pixels in the CAVE data set from different methods. Each pixel, along with its corresponding spectrum, is marked with a unique color. The scaling factor is set as 8, and the SNRs of both the LR HSI and HR MSI are 30 dB. (a) CAVE data set. (b) Yong. (c) NSSR. (d) CMS. (e) uSDN. (f) Ours.

large or small. Moreover, when $T = 1$, the pixel-aware weights together with the input of the generator network update every epoch. To balance the accuracy and the update speed, we set T as 100 in all experiments in this study.

3) Weight Coefficient α: The weight coefficient α influences the magnitude of weight $\mathbf{W}$. As in the above ablation study, we conduct this experiment on the CAVE data set and set the SR scale factor s as 8. To find the optimal α that yields the best performance, we evaluate the proposed method with different values of α. Without loss of generality, we vary α in the range of [1, 3] with an interval of 0.5. The experimental results are plotted in Fig. 8. As can be seen from the figure, the proposed method obtains the highest PSNR value when $\alpha = 2.5$. Although it is possible to utilize a smaller interval for α to locate a more accurate optimal value, the proposed method outperforms most existing state-of-the-art methods. In the interests of simplicity, we set $\alpha = 2.5$ empirically for the proposed method in all experiments.

TABLE V

PERFORMANCE OF EACH METHOD ON THE CAVE DATA SET WITH DIFFERENT SR SCALE FACTORS, WHILE THE LR HSI IS CONTAMINATED BY NOISE (WITH SNR OF 30 dB). THE BEST RESULTS ARE IN BOLD

SR Scale factor	Methods	RMSE	PSNR	SAM	SSIM
$s = 8$	Yong	5.81	37.84	7.60	0.9574
	NSSR	2.13	41.96	8.09	0.9700
	CMS	2.03	42.28	8.19	0.9712
	uSDN	4.21	36.48	15.26	0.9191
	Ours	**1.98**	**42.45**	**4.01**	**0.9867**
$s = 16$	Yong	3.25	38.37	7.59	0.9716
	NSSR	3.43	38.03	7.51	0.9703
	CMS	2.23	41.54	5.36	0.9830
	uSDN	4.60	35.53	15.80	0.9059
	Ours	**2.18**	**41.61**	**4.34**	**0.9862**
$s = 32$	Yong	4.33	36.01	9.17	0.9597
	NSSR	4.52	35.69	10.10	0.9696
	CMS	2.85	39.56	5.79	0.9815
	uSDN	6.07	33.63	18.95	0.8773
	Ours	**2.72**	**39.85**	**5.21**	**0.9829**

C. Performance Comparison on Different Data Sets

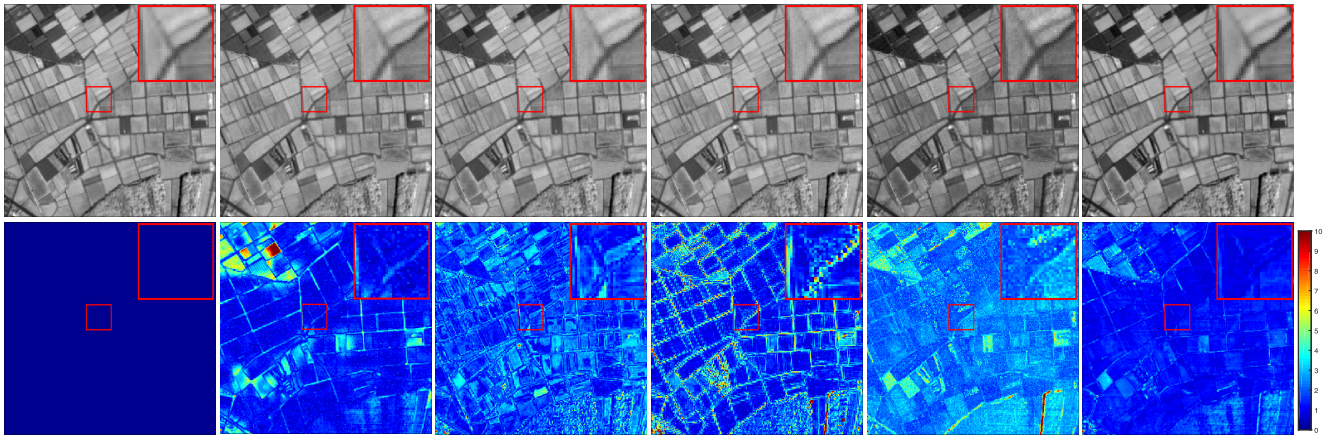
In this section, we compare the performance of the proposed method with all competing methods on different data sets.

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TABLE VI

PERFORMANCE OF EACH METHOD ON THE CHIKUSEI DATA SET WITH DIFFERENT NOISY OBSERVATIONS. THE SR SCALE FACTOR IS SET AS 8, AND THE BEST RESULTS ARE IN BOLD

Methods	HSI 30dB and MSI 0dB				HSI 0dB and MSI 30dB				HSI 30dB and MSI 30dB			
	RMES	PSNR	SAM	SSIM	RMES	PSNR	SAM	SSIM	RMES	PSNR	SAM	SSIM
Yong	1.65	43.60	3.47	0.9729	1.80	42.92	3.71	0.9680	1.79	42.89	3.66	0.9677
NSSR	2.61	39.78	4.05	0.9504	1.72	43.32	**3.08**	0.9768	2.57	39.91	4.06	0.9521
CMS	3.06	38.41	3.86	0.9412	2.98	38.67	3.24	0.9468	3.08	38.35	3.94	0.9399
uSDN	3.25	38.00	6.94	0.9215	3.06	38.39	5.79	0.9337	3.08	38.35	6.60	0.9291
Ours	**1.64**	**43.62**	**3.21**	**0.9786**	**1.69**	**43.43**	3.18	**0.9782**	**1.74**	**43.10**	**3.33**	**0.9759**



(a) (b) (c) (d) (e) (f)

Fig. 11. Reconstruction results of the Chikusei data. The first row contains the reconstructed 90th band images from different methods, while the second row presents the reconstruction error maps. The SR scale factor is set as 16, and the SNRs of both the LR HSI and HR MSI are 30 dB. (a) GT. (b) Yong. (c) NSSR. (d) CMS. (e) uSDN. (f) Ours.

TABLE VII

PERFORMANCE OF EACH METHOD ON THE CHIKUSEI DATA SET WITH DIFFERENT SR SCALE FACTORS, IN WHICH THE LR HSI AND HR MSI ARE BOTH CONTAMINATED BY NOISE (WITH SNRS OF 30 dB). THE BEST RESULTS ARE IN BOLD

SR Scale factor	Methods	RMSE	PSNR	SAM	SSIM
$s = 8$	Yong	1.79	42.89	3.66	0.9677
	NSSR	2.57	39.91	4.06	0.9521
	CMS	3.08	38.35	3.94	0.9399
	uSDN	3.08	38.35	6.60	0.9291
	Ours	**1.74**	**43.10**	**3.33**	**0.9759**
$s = 16$	Yong	2.29	40.81	4.71	0.9628
	NSSR	2.69	39.52	4.58	0.9500
	CMS	3.48	37.40	4.09	0.9353
	uSDN	3.40	37.49	6.93	0.9375
	Ours	**1.88**	**42.50**	**3.73**	**0.9741**
$s = 32$	Yong	2.59	39.81	5.30	0.9542
	NSSR	2.18	41.31	4.26	0.9611
	CMS	4.98	34.30	5.94	0.9010
	uSDN	4.58	34.86	8.77	0.9045
	Ours	**2.07**	**41.67**	**4.05**	**0.9702**

1) Experiments on CAVE Data Set: To testify the effectiveness of the proposed method, we conduct sufficient experiments on the CAVE data set, including experiments with different SR scale factors and experiments with different levels of noise.

a) Noisy LR HSI: In this experiment, we set the SR scale factor as 8 and reconstruct the HR HSI using a noisy LR HSI and a clean HR MSI. To determine how the performance of all methods is influenced by noise, different intensities of noise are added separately into the LR HSI, which results in the SNR of the generated noisy LR HSI being 30, 25, and 20 dB, respectively. It is notable that a larger intensity of noise results in a smaller SNR value of the generated noisy LR HSI. The numerical results on the CAVE data set are summarized in Table II, in which the results are obtained by averaging the reconstruction results of each HSI in the CAVE data set. From the results, we can see that Ours obtains the best reconstruction results at any level of noise intensity. Although the performance of all methods degrades when noise intensity increases (i.e., from 30 to 20 dB), the performance of Ours is more stable than that of the competing methods, which demonstrates the robustness of the proposed method to noise.

b) Noisy HR MSI: In this experiment, we reconstruct the HR HSI using a clean LR HSI and a noisy HR MSI. Similar to the settings used in the abovementioned experiment, we set the SR scale factor as 8 and alter the intensity of noise in the HR MSI, which makes the SNR of the generated HR MSI 30, 25, and 20 dB, respectively. The numerical results on the CAVE data set are reported in Table III. We can see from the results that the proposed method achieves the best performance in terms of both reconstruction accuracy and stability compared with other methods. By comparing the experimental results

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TABLE VIII

PERFORMANCE OF EACH METHOD ON THE WDC DATA SET WITH DIFFERENT NOISY OBSERVATIONS. THE SR SCALE FACTOR IS SET AS 8, AND THE BEST RESULTS ARE IN BOLD

Methods	HSI 30dB and MSI 0dB				HSI 0dB and MSI 30dB				HSI 30dB and MSI 30dB			
	RMES	PSNR	SAM	SSIM	RMES	PSNR	SAM	SSIM	RMES	PSNR	SAM	SSIM
Yong	4.84	34.39	3.55	0.8908	4.98	34.14	3.63	0.8886	4.90	34.28	3.58	0.8893
NSSR	1.97	42.03	1.60	0.9793	1.74	43.03	1.25	0.9837	1.93	42.18	1.58	0.9804
CMS	2.67	39.49	2.18	0.9662	2.26	40.91	1.60	0.9754	2.67	39.47	2.19	0.9664
uSDN	3.29	37.72	1.96	0.9833	3.53	37.10	2.00	0.9806	3.29	37.72	1.88	0.9820
Ours	**1.75**	**42.92**	**1.34**	**0.9862**	**1.55**	**44.04**	**1.12**	**0.9899**	**1.72**	**43.17**	**1.36**	**0.9864**

TABLE IX

PERFORMANCE OF EACH METHOD ON THE WDC DATA SET WITH DIFFERENT SR SCALE FACTORS, IN WHICH THE LR HSI AND HR MSI ARE BOTH CONTAMINATED BY NOISE (WITH SNRS OF 30 dB). THE BEST RESULTS ARE IN BOLD

SR Scale factor	Methods	RMSE	PSNR	SAM	SSIM
$s = 8$	Yong	4.90	34.28	3.58	0.8893
	NSSR	1.93	42.18	1.58	0.9804
	CMS	2.67	39.47	2.19	0.9664
	uSDN	3.29	37.72	1.88	0.9820
	Ours	**1.72**	**43.17**	**1.36**	**0.9864**
$s = 16$	Yong	5.92	32.65	4.42	0.8802
	NSSR	4.68	34.68	3.82	0.9275
	CMS	3.01	38.45	2.26	0.9651
	uSDN	2.92	38.71	1.90	0.9825
	Ours	**2.17**	**41.22**	**1.66**	**0.9829**
$s = 32$	Yong	6.88	31.35	5.43	0.8697
	NSSR	5.38	33.48	4.59	0.9361
	CMS	4.34	35.34	3.30	0.9496
	uSDN	4.90	34.29	3.86	0.9709
	Ours	**2.39**	**40.41**	**1.85**	**0.9806**

from the noisy HR MSI with that from the noisy LR HSI, we can conclude that SAM from the former is better than that from the latter. This conclusion is reasonable since LR HSI provides the primary spectral information for reconstruction. Once the LR HSI is corrupted by noise, the reconstruction performance will be more affected.

c) Noisy LR HSI and noisy HR MSI: In this experiment, we consider a more complex situation in which both LR HSI and HR MSI are contaminated by noise simultaneously. Similar to the abovementioned experiments, we set the SR scale factor as 8 and alter the intensity of noise in HR MSI and LR HSI. The corresponding numerical results are summarized in Table IV. The results are consistent with those of the abovementioned experiments, i.e., the proposed method has obvious superiority over other competing methods.

To further clarify the superiority of the proposed method, the SR results of one example HSI from the CAVE data set ("chart_and_stuffed_toy") from noisy observations with 30-dB SNR are visualized in Fig. 9. In this figure, the first row gives the reconstructed 31st band images of the example HSI from different methods, while the second row lists the corresponding error maps. Within these images, we zoom into a local region (enclosed by the red square) to further observe

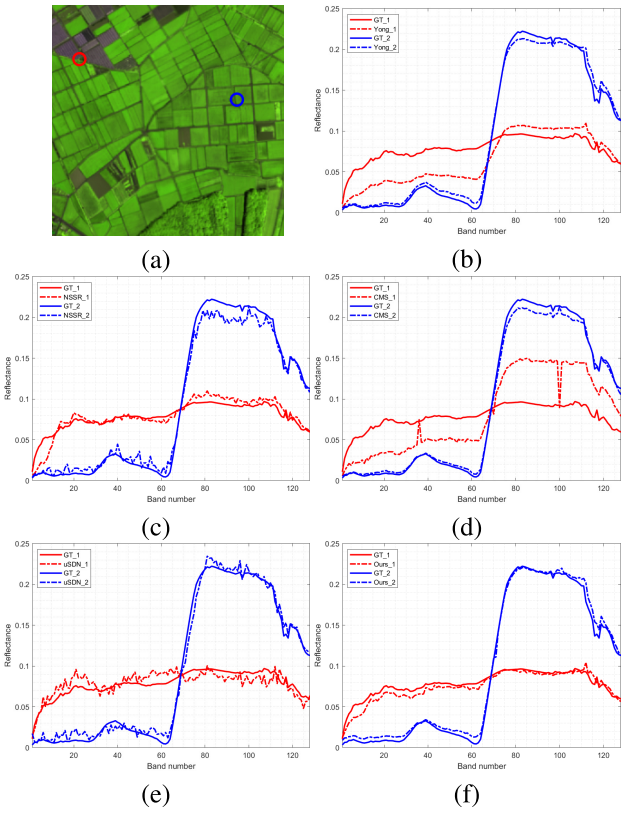


Fig. 12. Reconstructed spectral curve of two representative pixels in the Chikusei data set from different methods. Each pixel, along with its corresponding spectrum, is marked with a unique color. The SR scale factor is set as 16, and the SNRs of both the LR HSI and HR MSI are 30 dB. (a) Chikusei data set. (b) Yong. (c) NSSR. (d) CMS. (e) uSDN. (f) Ours.

the details. Since a darker color on the error map indicates better reconstruction results, it is evident that the proposed method can competently reconstruct the HR HSI from noisy LR HSI and HR MSI, as it recovers more details of the structures and produces a smaller error compared with other competing methods. In addition, the reconstructed spectral curves of two representative pixels are also visualized in Fig. 10, in which each pixel and its corresponding spectrum are marked with a unique color. To facilitate comparison, the reconstructed spectrum is plotted with a dotted line, while the ground-truth spectrum is plotted with a solid line. From Fig. 10, it is evident that the proposed method accurately reconstructs the spectra, which is consistent with the numerical results reported in Table IV.

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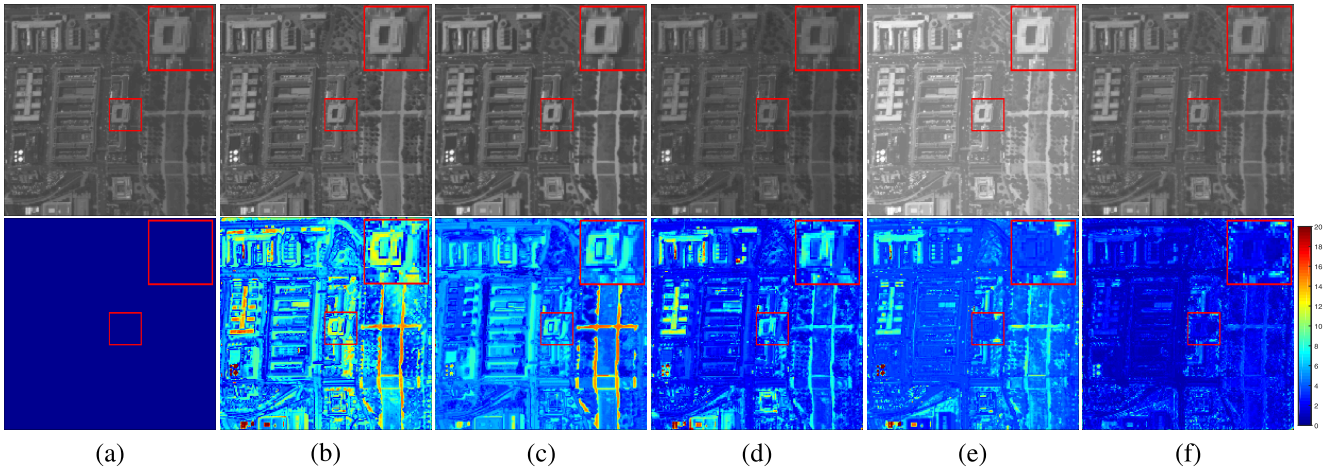


Fig. 13. Reconstruction results of the Wdc data. The first row gives the reconstructed 20th band images from different methods, while the second row presents the reconstruction error maps. The SR scale factor is set as 32, and both SNRs of LR HSI and HR MSI are 30 dB. (a) GT. (b) Yong. (c) NSSR. (d) CMS. (e) uSDN. (f) Ours.

d) Performance with different SR scale factors: In this experiment, we testify the applicability of all methods at different SR scale factors. More specifically, we fix the SNR of the observed LR HSI to 30 dB and then reconstruct the HR HSI by setting the SR scale factor s as 8, 16, and 32, respectively. The numerical results on the CAVE data set are summarized in Table V. It can be seen that a larger SR scale factor enhances the difficulty for HSI SR, which results in the deterioration of all methods when the SR scale factor is increased. Nevertheless, the proposed method obtains the best and most stable reconstruction results among all methods.

2) Experiments on the Chikusei Data Set: In this section, two experiments are conducted on the Chikusei data set. First, we set the SR scale factor as 8 and conduct HSI SR with different levels of noise to verify the performance of all methods on the Chikusei data set. More specifically, we reconstruct the HR HSI via noisy LR HSI (with clean HR MSI), noisy HR MSI (with clean LR HSI), and both noisy LR HSI and noisy HR MSI. The SNR is set as 30 dB for those noisy images. The numerical results are summarized in Table VI. From this table, we can see that the proposed method has obvious advantages over other competing methods given any noisy observations.

Subsequently, using a noisy LR HSI and noisy HR MSI (the SNRs of which are 30 dB), we reconstruct the HR HSI by setting the SR scale factor as 8, 16, and 32, respectively. We summarize the numerical results in Table VII. In addition, the SR results of the Chikusei data, reconstructed by different methods when the SR scale factor is set as 16, are visualized in Fig. 11. Both the representative reconstructed band image and the average reconstruction error map produced by the proposed method are the closest to the ground truth. Accordingly, after assessing both quantitative measures and visual results, we can see that the proposed method has marked advantages when reconstructing the spatial information of an HSI. To further verify the reconstruction performance on spectra, we choose two pixels and plot their spectra (both that obtained from the ground truth image and that obtained from reconstructed image) in Fig. 12. We can see that the proposed method obtains the best spectral reconstruction results that are most similar to the ground truth.

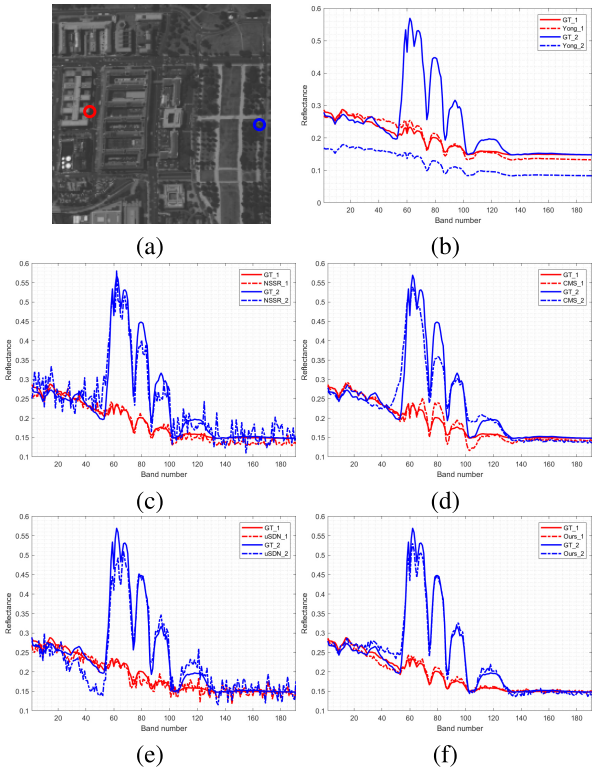


Fig. 14. Reconstructed spectral curve of two representative pixels in the Wdc data set from different methods. Each pixel, along with its corresponding spectrum, is marked with a unique color. The SR scale factor is set as 32, and the SNRs of both the LR HSI and HR MSI are 30 dB. (a) Washington DC data set. (b) Yong. (c) NSSR. (d) CMS. (e) uSDN. (f) Ours.

3) Experiment on the Wdc Data Set: To testify the applicability of the proposed method to different kinds of data sets, we further conduct the following experiment on the Wdc data set. Using similar settings to those applied on the Chikusei data set, we first conduct HSI SR from different noisy observations with SR scale factor $s = 8$ and then testify all methods varied with different SR scale factors providing noisy HR MSI and LR HSI. We summarize the results from these two

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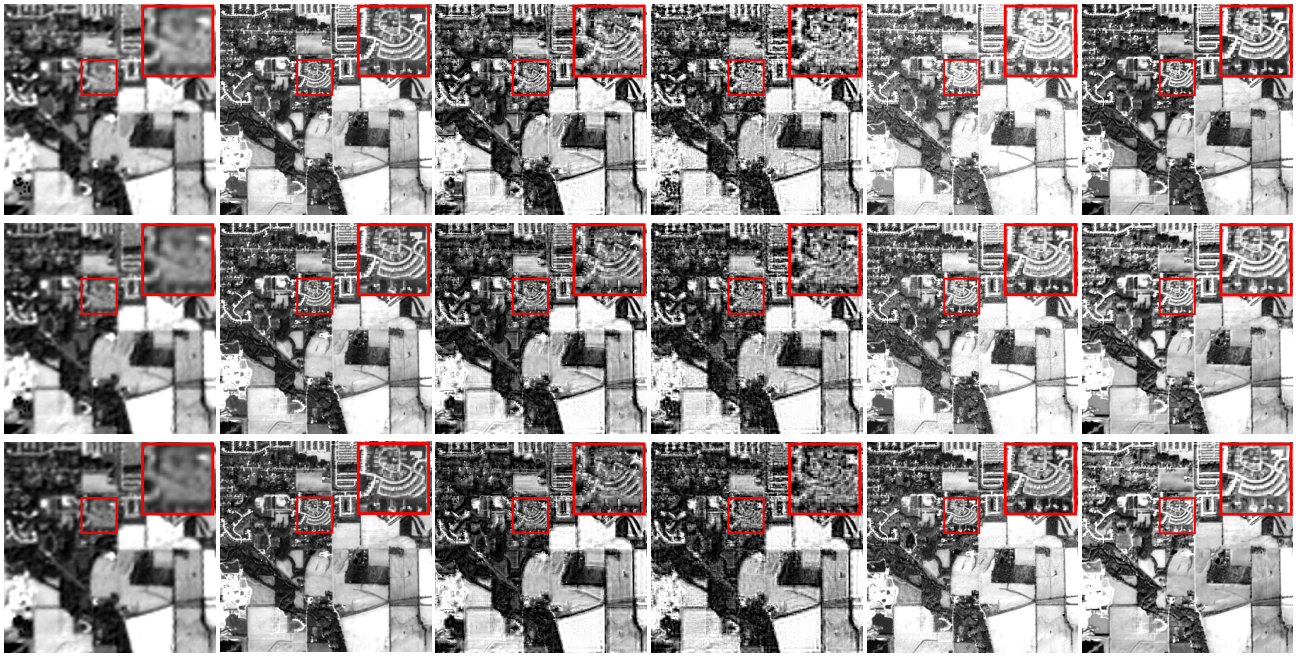


Fig. 15. Reconstructed results of all methods on the HypSen data set. From the first row to the third row, we list the seventh, 12th, and 15th bands of the reconstructed results, respectively. Bicubic denotes the reconstruction result obtained via bicubic interpolation. (a) Bicubic. (b) Yong. (c) NSSR. (d) CMS. (e) uSDN. (f) Ours.

experiments in Tables VIII and IX, respectively. Visualization results of the reconstructed HSIs in the Wdc data set from different methods when the SR scale factor is set as 32 are provided in Fig. 13, where the SNRs of both the LR HSI and HR MSI are 30 dB. It can be seen that the proposed method has obvious advantages over other competing methods. This conclusion is also valid for the spectral reconstruction, as can be seen from Fig. 14. It is noticeable that, although some methods, such as Yong, perform well on the Chikusei data set, their performance drops significantly on the Wdc data set. By contrast, the proposed method achieves stable performance on all data sets, including CAVE, Chikusei, and Wdc, which demonstrates that the proposed method can be applied to different HSI data sets.

4) Experiment on Real Data: To further verify the proposed method on a real data set, we conduct the HSI SR experiments on the HypSen data set. More specifically, we treat the HSI and MSI data in HypSen as the LR HSI X and HR MSI Y and then use them to reconstruct the HR HSI and compare the results obtained from different methods. In this experiment, the SR scale factor is set as 3 since the spatial resolutions of the adopted LR HSI and HR MSI are 100×100 and 300×300, respectively. To facilitate direct comparison, we also utilize a bicubic interpolation method to reconstruct the HSI (termed Bicubic in this study). The reconstructed results are shown in Fig. 15, in which three representative bands are chosen for visual comparison. Within these images, we still zoom in on a local region (enclosed by the red square) to further observe the details. It can be seen that Bicubic fails to reconstruct the HR HSI. The images obtained from both CMS and NSSR are blurred, and the details within these images cannot be distinguished. For Yong, the resulting images are oversmoothed. By contrast, the proposed method recovers more details of the structures and obtains the best SR results. In addition, we also evaluate the SR performance of all methods by means of a nonreference image quality evaluation metric [45] and summarize the corresponding numerical results in Table X (in which a smaller value represents better SR performance).

TABLE X

NO-REFERENCE HSI QUALITY MEASUREMENT SCORE OF EACH METHOD ON THE REAL HSI SR TASK (THE SMALLER THE VALUE, THE BETTER)

Methods	Bicubic	Yong	NSSR	CMS	uSDN	Ours
Score	31.01	12.47	12.84	12.83	12.86	**12.40**

From the abovementioned experimental results on these four data sets, we can conclude that the proposed method can obtain better SR performance than other state-of-the-art competing methods providing noisy observation images. We attribute the performance degradation of other comparison methods (i.e., Yong, NSSR, and CMS) into the heuristic handcrafted image prior they utilized, which shows limited capacity in capturing specific characteristics of the HSI, as well as generalizing to complex experiment settings (e.g., noisy observation images and large SR scale factor).

V. CONCLUSION

In this study, we focus on the unsupervised fusion-based HSI method to reconstruct HR HSI from observed LR HSI and conventional HR image. Rather than adopting the

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heuristic handcrafted image prior, we first investigate the exploitation of a deep image prior constructed by a deep neural network to capture the specific statistics of images. By doing this, the proposed method builds a connection between deep learning-based methods and conventional traditional optimization-based methods. Within this framework, we further utilize the intermediate reconstruction results to guide the unsupervised learning and, consequently, propose a recurrent fusion-based HSI SR method using pixel-aware refinement, which has improved capacity for fusion-based HSI SR. In contrast with the state-of-the-art methods, the proposed method not only presents a new fusion-based HSI SR framework but also exhibits strong robustness over different intensities of noise corruption. Extensive experiments on four HSI data sets demonstrate the effectiveness of the proposed method for HSI SR, in both accuracy and robustness, compared with other existing state-of-the-art methods.

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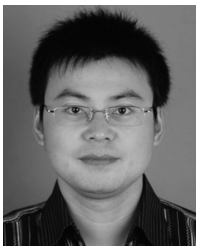
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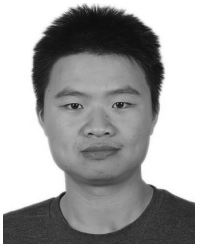
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